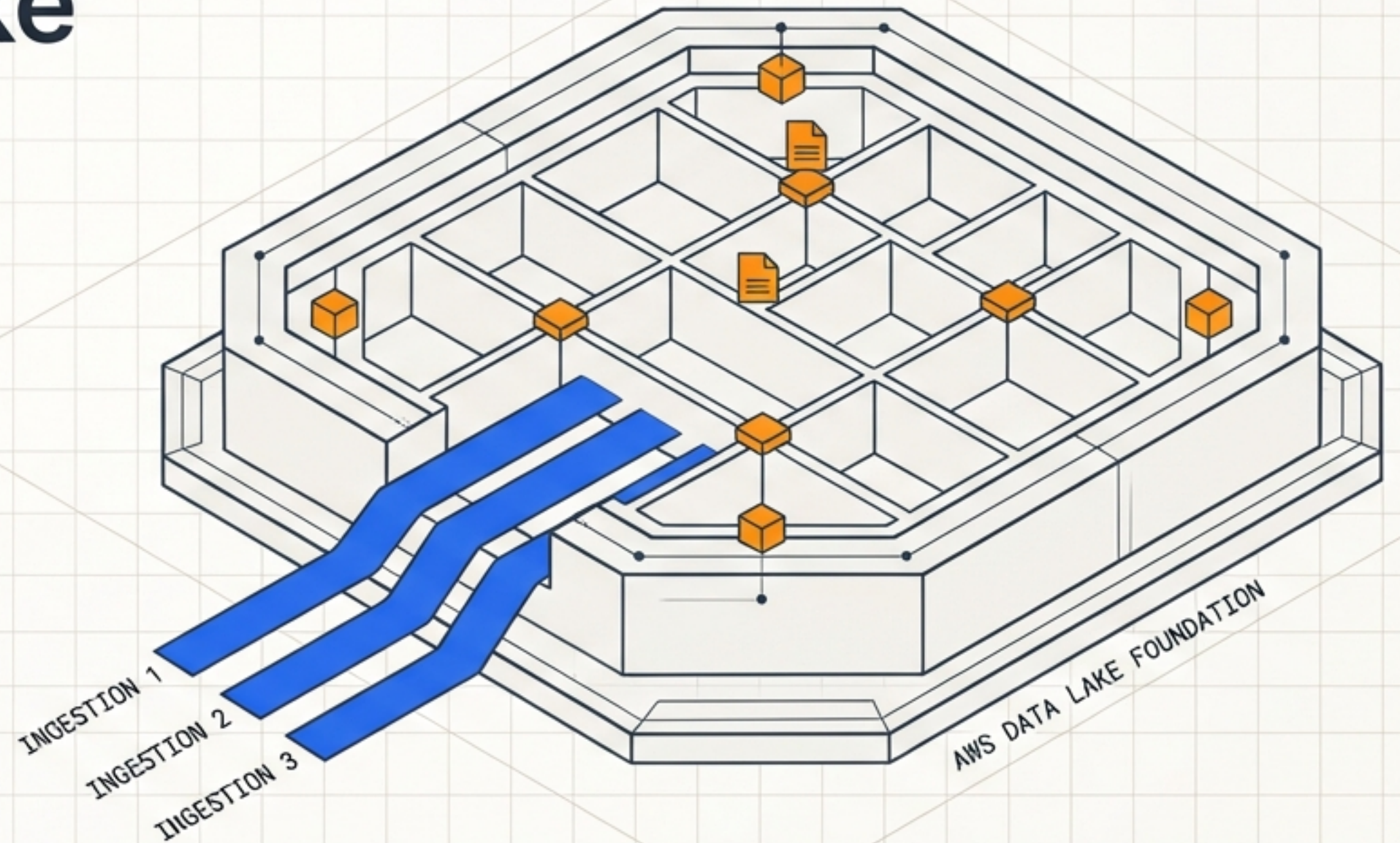


Architecting the Modern Data Lake

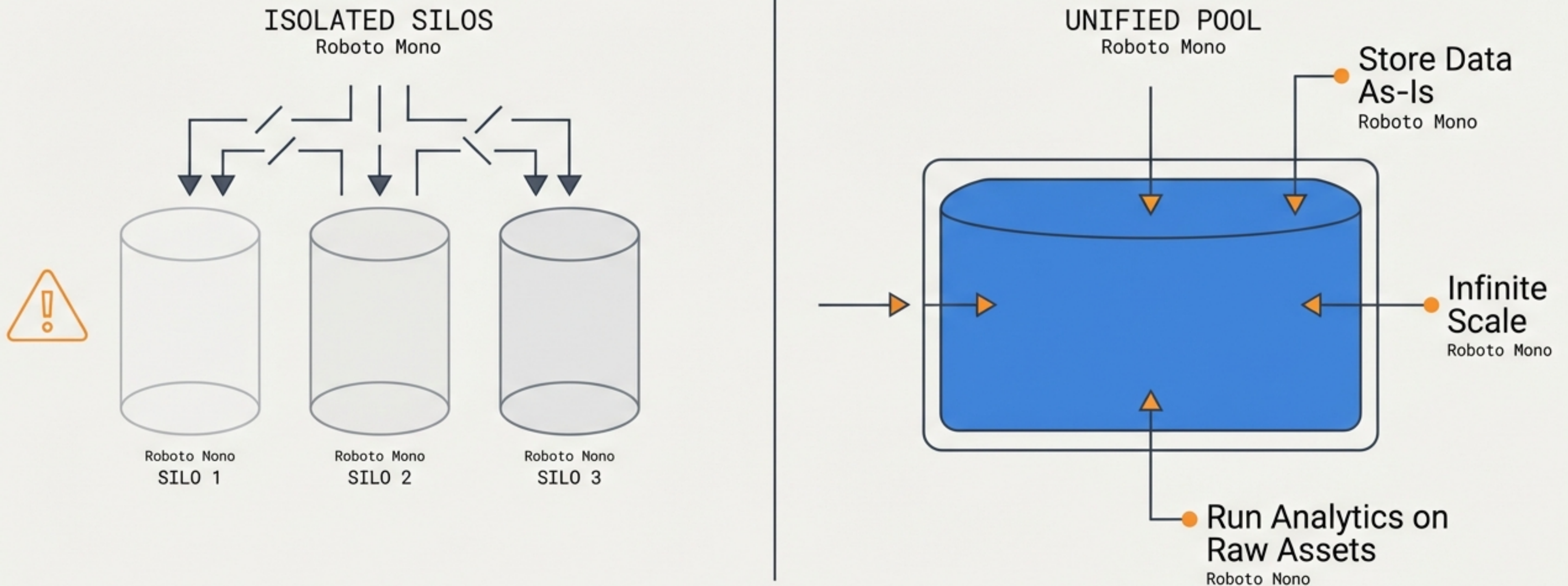
A comprehensive guide to building, securing, and scaling data ecosystems on AWS.



Based on the highly-rated Coursera AWS Cloud Solutions Architect Professional Certificate program.

Breaking Down Data Silos

Enterprise data is massive, messy, and siloed. A data lake provides a centralized repository that allows you to store all your structured, semi-structured, and unstructured data at any scale.



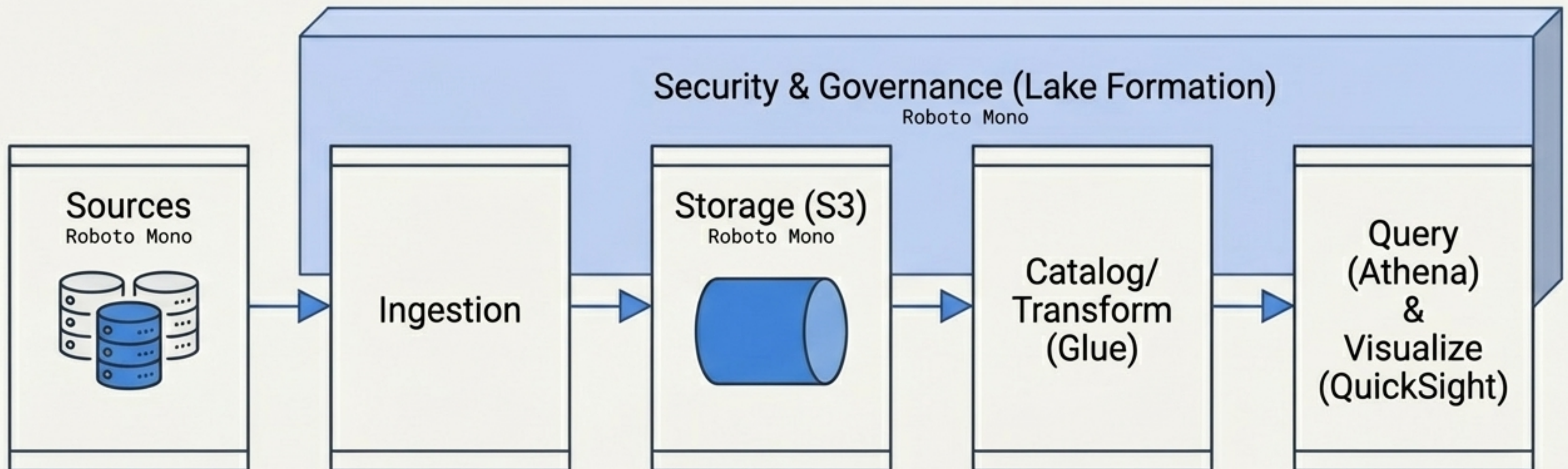
Data Lake vs. Data Warehouse

Metric	Data Lake	Data Warehouse
Data Types	All Data	Relational/Structured
Schema	Schema-on-Read / Applied during analysis	Schema-on-Write / Applied before storage
Storage Cost	Low cost / S3 architecture	Higher cost / Compute-coupled
Primary Users	Data Scientists & Engineers	Business Analysts

You don't replace the warehouse; you build a data lake to feed and complement it.

The AWS Data Lake Ecosystem Blueprint

Over 90 fully-featured services power the AWS ecosystem. Building a modern data lake requires mastering this precise combination.



The Masterclass: Building Data Lakes on AWS

Instructors

Taught by AWS Experts Rafael Lopes & Alex G.
Combined 360k+ learners

Rating

4.7 / 5 Stars
Based on 308 reviews (92% positive rating)

Commitment

1 week @ 10 hours/week
Flexible, self-paced

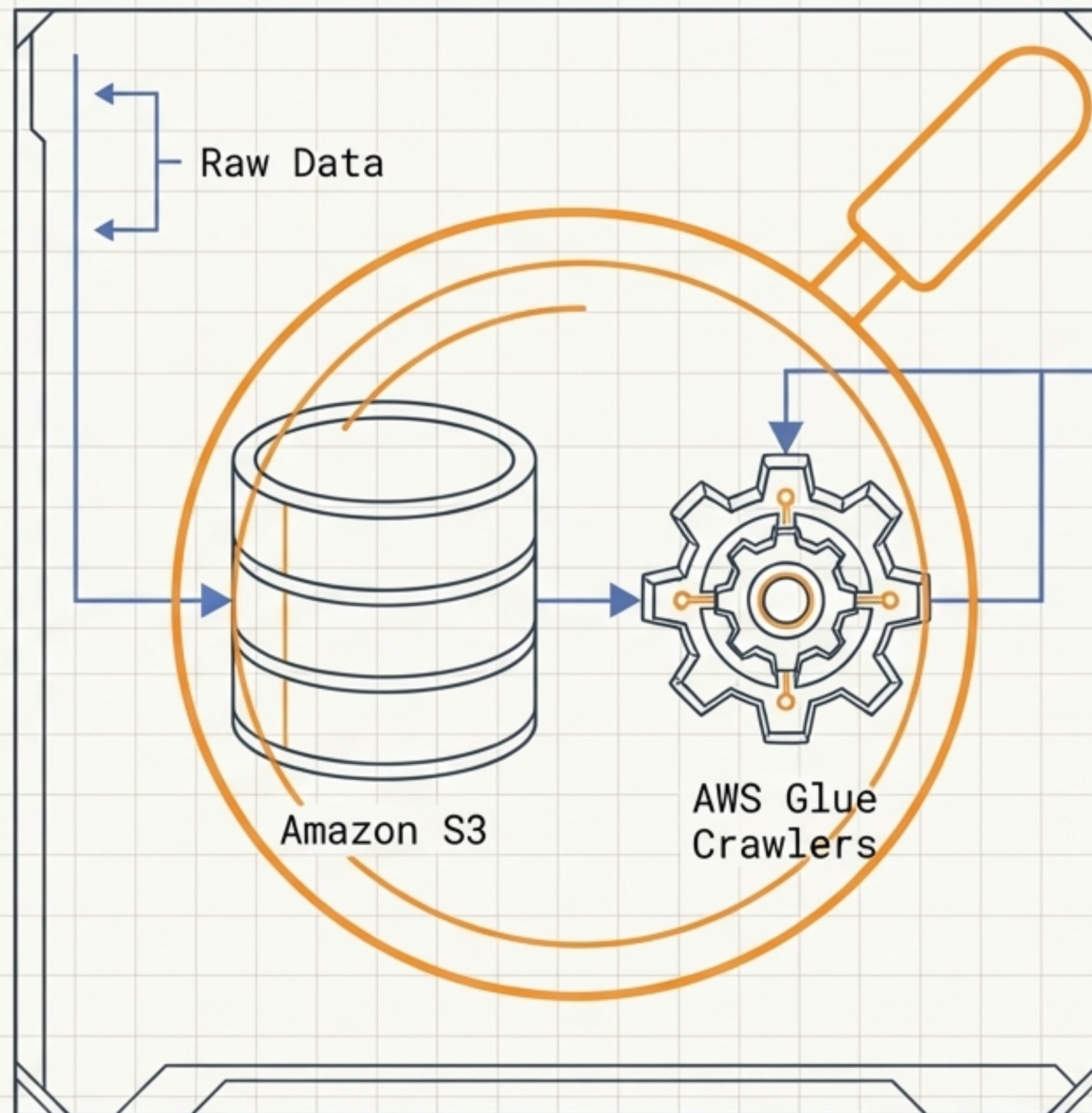
Prerequisite

Beginner friendly
Requires AWS Cloud Technical Essentials

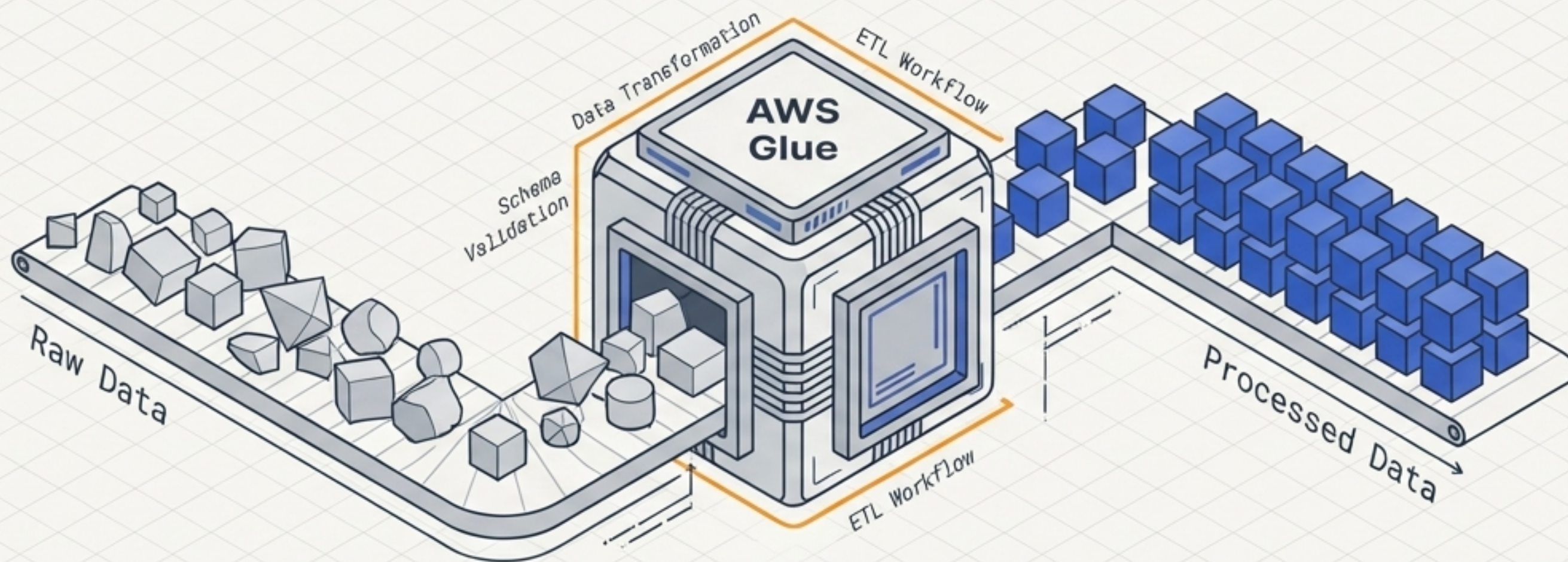
A 6-module journey to architecting the exact blueprint on the previous slide.

Phase 1: Ingestion, Cataloging, & Storage

- 1 The Bedrock (Amazon S3)**
Ingesting massive data streams into low-cost, scalable storage.
- 2 The Magic of Cataloging**
AWS Glue Crawlers automatically scan raw data, discover file formats, and infer the schema without manual mapping.
- 3 Optimization**
Formatting, partitioning, and compressing data layers for high-speed, cost-effective retrieval.



Phase 2: Transformation & Processing



Serverless Integration ■

Using AWS Glue for zero-infrastructure data integration and processing.

Visual ETL ■

Leveraging Glue Studio and Glue Databrew to clean and prepare data using visual, code-free interfaces.

Automation ■

Orchestrating jobs and workflows to keep the data pipeline moving automatically.

Phase 3: The Governance Shield

Securing chaotic raw data is notoriously difficult. AWS Lake Formation makes it manageable.

Automated Creation

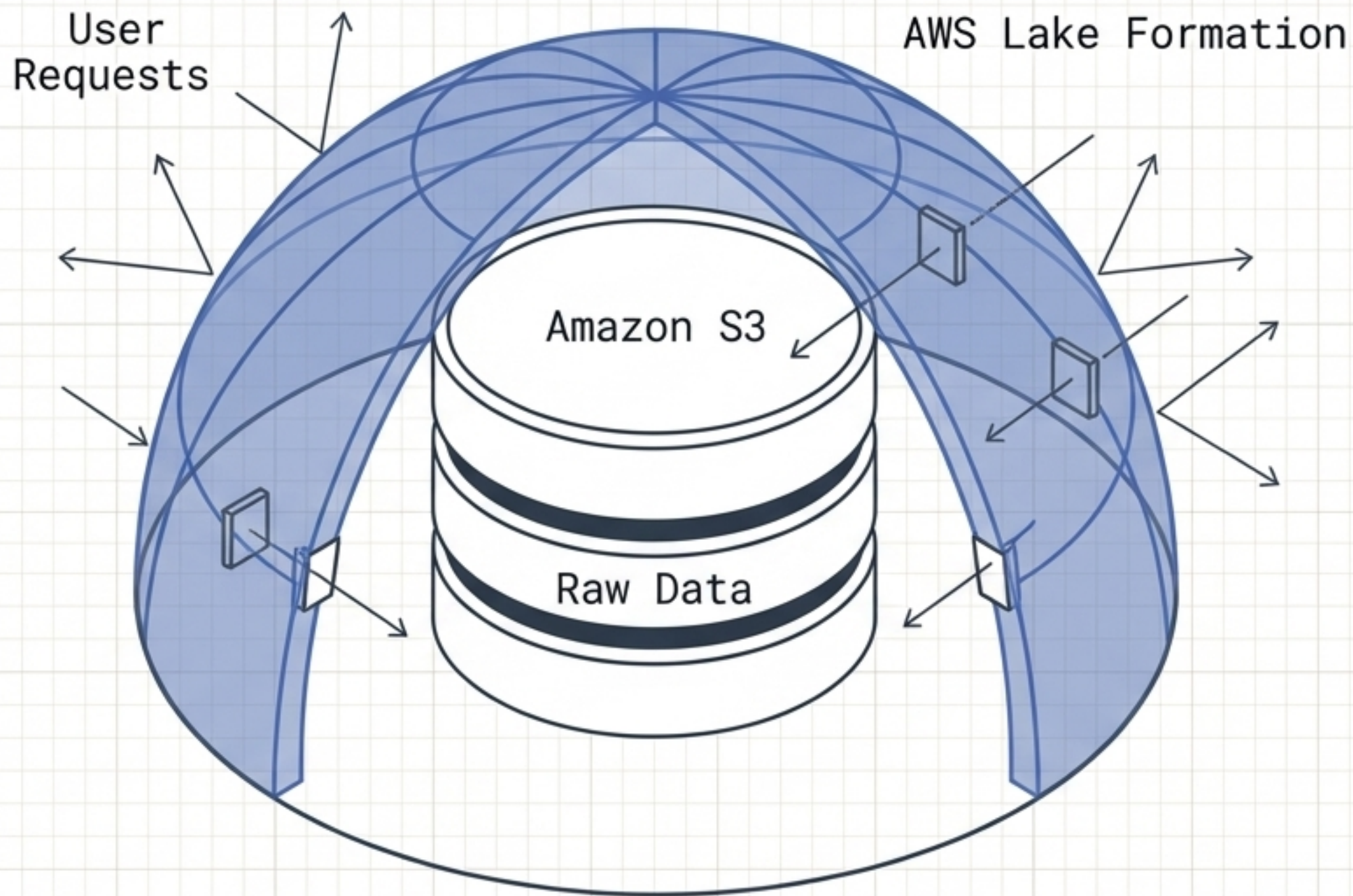
Spin up data lakes instantly using Lake Formation Blueprints.

The Permission Model

Moving beyond complex IAM policies to centralized, simple Lake Formation grants.

Fine-Grained Access

Securing data down to the specific row and column level using simple LF-Tags.



Phase 4: Querying & Visualization



Amazon Athena

Serverless, interactive querying.
Run standard SQL directly against
S3 data. No loading required.

Federated Queries

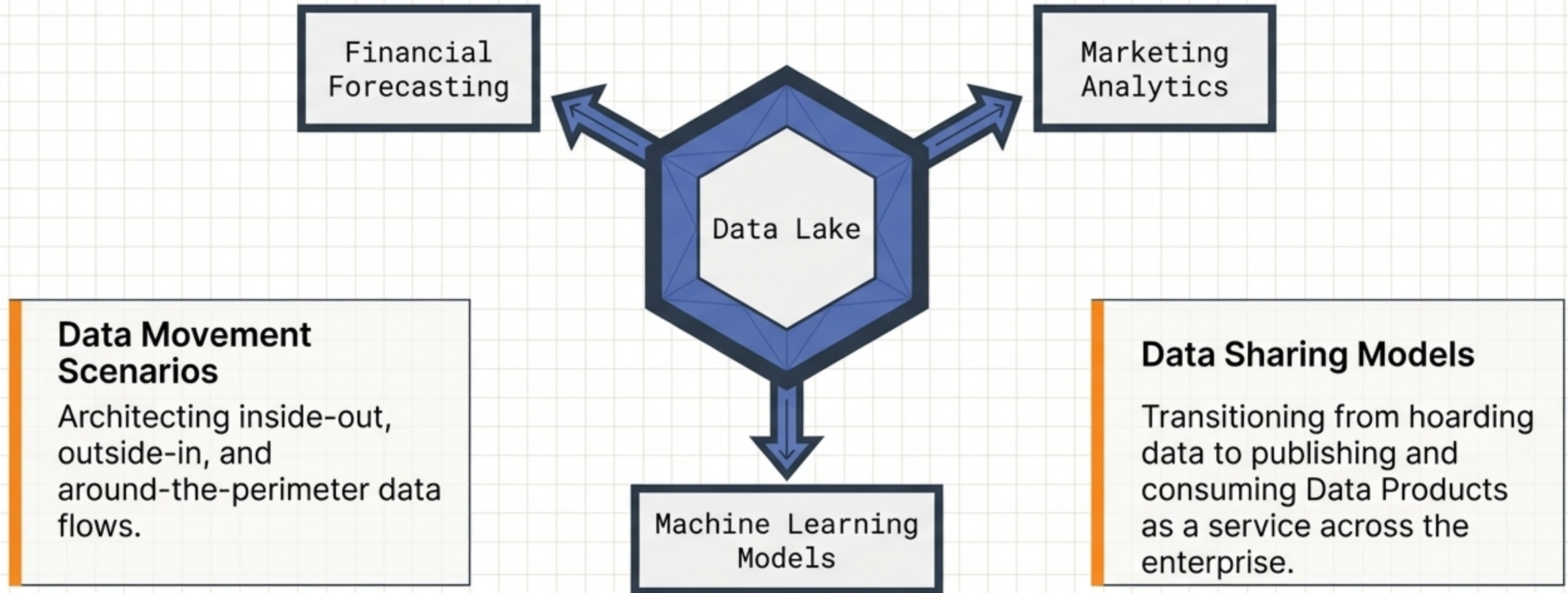
Query data wherever it lives
across your architecture.

Amazon QuickSight

Bring the raw data to life through
interactive, enterprise-grade
dashboards.

Synthesis: The Modern Data Architecture

A successful data lake is not a stagnant pool; it is an active exchange.



From Theory to Practice: Hands-On Labs

3 hours of applied, real-world AWS console experience.

Lab 1 (60 min)

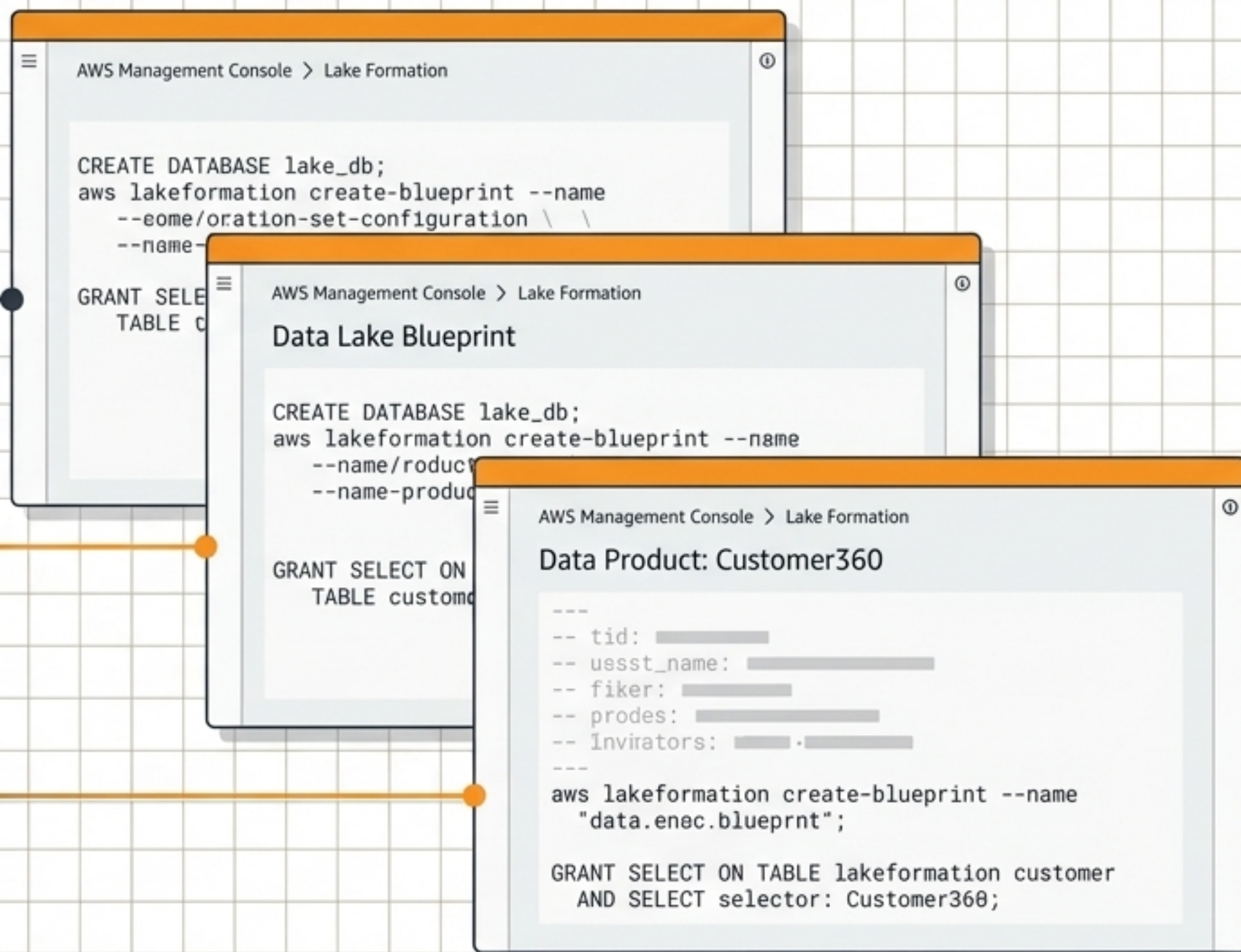
Building a Data Lake from scratch using AWS Lake Formation.

Lab 2 (60 min)

Automating secure Data Lake creation using Blueprints.

Lab 3 (60 min)

Publishing and Managing Data Products in Lake Formation.



Learner Outcomes & Career Trajectory

Skills Acquired

Data Architecture

Data Engineering

Machine Learning

Data Governance



I directly applied the concepts and skills I learned from my courses to an exciting new project at work.

– Jennifer J.



The best AWS course... taught by two excellent instructors who use relevant examples, demonstrations, graphics, and content that is carefully organized.

– Reviewer S.T.

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